

The Argument of a Complex Number

The Geometric and Algebraic Definition

A nonzero complex number $z = x + iy$ determines a point in the complex plane. The argument of z , written $\arg(z)$, is the angle φ measured in radians from the positive real axis to the ray from the origin through z , taken positive in the counterclockwise direction.

Equivalently, φ is any real number satisfying

$$z = r(\cos \varphi + i \sin \varphi) = r e^{i\varphi}, \quad (1)$$

where

$$r = |z| = \sqrt{x^2 + y^2} > 0 \quad (2)$$

is the modulus of z . The pair (r, φ) constitutes the polar representation of z .

The argument of zero is left undefined: there is no ray from the origin to the origin, and (1) degenerates to $0 = 0$ for every φ .

Why the argument is not a single number

Equation (1) constrains φ only through $\cos \varphi$ and $\sin \varphi$. Because both functions are 2π -periodic, if φ_0 is one solution of (1), then so is $\varphi_0 + 2\pi n$ for every integer n . Geometrically: rotating the ray by any whole number of full turns lands on the same point. So the argument of a nonzero complex number is not a single real number but an infinite discrete set,

$$\arg(z) = \{\varphi_0 + 2\pi n \mid n \in \mathbb{Z}\}, \quad (3)$$

where φ_0 is any one value satisfying (1). This makes $\arg$ a multi-valued function on $\mathbb{C} \setminus \{0\}$.

When one writes " $\arg(z) = \theta$ " without further qualification, one is asserting that θ is some element of the set (3) — the particular element is usually clear from context or convention.

The Principal Value

To extract a single-valued function from the multi-valued $\arg$, one selects exactly one representative from each coset (3). The standard convention is the **principal value**, denoted $\text{Arg}(z)$, defined as the unique $\varphi \in (-\pi, \pi]$ satisfying (1). That is,

$$\text{Arg}(z) \in (-\pi, \pi], \quad z = |z| e^{i \text{Arg}(z)}. \quad (4)$$

This means:

Location of z	$\text{Arg}(z)$
Positive real axis ($x > 0, y = 0$)	0
Upper half-plane ($y > 0$)	$\in (0, \pi)$
Negative real axis ($x < 0, y = 0$)	π
Lower half-plane ($y < 0$)	$\in (-\pi, 0)$

The choice $(-\pi, \pi]$ is not the only possibility. Some authors use $[0, 2\pi)$. What matters is that the interval has length 2π and is half-open, so exactly one representative is chosen from each equivalence class modulo 2π .

Computing the principal value from Cartesian coordinates

Given $z = x + iy$, the principal value is computed by a quadrant-aware extension of the arctangent. The ordinary $\arctan(y/x)$ has range $(-\pi/2, \pi/2)$ and cannot distinguish (x, y) from $(-x, -y)$, since both yield the same ratio. To recover the full angle in $(-\pi, \pi]$, one must use the signs of x and y independently:

$$\text{Arg}(x + iy) = \begin{cases} \arctan\left(\frac{y}{x}\right) & x > 0, \\ \arctan\left(\frac{y}{x}\right) + \pi & x < 0, y \geq 0, \\ \arctan\left(\frac{y}{x}\right) - \pi & x < 0, y < 0, \\ +\frac{\pi}{2} & x = 0, y > 0, \\ -\frac{\pi}{2} & x = 0, y < 0. \end{cases} \quad (5)$$

The corrections $\pm\pi$ in the second and third cases reflect the fact that when $x < 0$, the point lies in the second or third quadrant, and $\arctan(y/x)$ erroneously places it in the first or fourth. The signs of x and y individually determine which correction to apply.

An equivalent expression that avoids case analysis (except on the nonpositive real axis) is

$$\text{Arg}(x + iy) = 2 \arctan \frac{y}{\sqrt{x^2 + y^2} + x}, \quad (6)$$

valid whenever the denominator is nonzero, i.e., everywhere except when $y = 0$ and $x \leq 0$.

The Modulus-Argument Form and Algebraic Identities

Every nonzero complex number admits the modulus-argument form

$$z = |z| e^{i \text{Arg}(z)}, \quad (8)$$

which is simply a restatement of (4). From the multiplication rule $e^{i\alpha} e^{i\beta} = e^{i(\alpha+\beta)}$, the argument converts multiplication to addition and division to subtraction — but only modulo 2π , because the sum or difference may fall outside $(-\pi, \pi]$.

For nonzero z_1, z_2 :

$$\text{Arg}(z_1 z_2) \equiv \text{Arg}(z_1) + \text{Arg}(z_2) \pmod{2\pi}, \quad (9)$$

$$\text{Arg}\left(\frac{z_1}{z_2}\right) \equiv \text{Arg}(z_1) - \text{Arg}(z_2) \pmod{2\pi}, \quad (10)$$

$$\text{Arg}(z^n) \equiv n \text{Arg}(z) \pmod{2\pi}, \quad n \in \mathbb{Z}. \quad (11)$$

The notation $\equiv \pmod{2\pi}$ means: the two sides differ by an integer multiple of 2π , and the principal value is obtained by adding or subtracting the appropriate multiple to land in $(-\pi, \pi]$.

An explicit example

$$\text{Arg}\left(\frac{-1-i}{i}\right) = \text{Arg}(-1-i) - \text{Arg}(i) = -\frac{3\pi}{4} - \frac{\pi}{2} = -\frac{5\pi}{4}.$$

Since $-\frac{5\pi}{4} \notin (-\pi, \pi]$, the principal value is $-\frac{5\pi}{4} + 2\pi = \frac{3\pi}{4}$. Indeed, $\frac{-1-i}{i} = \frac{(-1-i)(-i)}{(-i)(i)} = \frac{i+i^2}{1} = -1+i$, which lies in the second quadrant at angle $\frac{3\pi}{4}$.

Connection to the complex logarithm

From $z = |z| e^{i \text{Arg}(z)}$, taking the principal logarithm gives

$$\text{Ln}(z) = \ln |z| + i \text{Arg}(z), \quad (12)$$

so $\text{Arg}(z) = \text{Im}(\text{Ln}(z))$. More generally, for any branch of the logarithm, $\arg(z) = \text{Im}(\log z)$. This is why the multi-valuedness of $\arg$ and $\log$ are manifestations of the same phenomenon.

Branch Cuts and Discontinuity

The principal value Arg is continuous on $\mathbb{C} \setminus (-\infty, 0]$ — the complex plane with the nonpositive real axis removed. On the negative real axis itself, Arg is discontinuous: approaching a point $x < 0$ from above gives argument tending to $+\pi$, while approaching from below gives argument tending to $-\pi$. At the point itself, the convention assigns the value π .

This ray $(-\infty, 0]$ is the branch cut of Arg . It is the price paid for forcing a single-valued determination out of a multi-valued function.

The branch cut is a choice, not a fact about the plane

There is nothing intrinsic to the negative real axis that forces the discontinuity there. If one instead defines the principal argument to lie in $[0, 2\pi)$, the branch cut moves to the positive real axis. More generally, for any $\alpha \in \mathbb{R}$, one can define

$$\arg_{\alpha}(z) \in [\alpha, \alpha + 2\pi), \quad (13)$$

which places the branch cut along the ray at angle α from the origin. The standard choice $\alpha = -\pi$ (giving $(-\pi, \pi]$) places the cut on the negative real axis. Every such choice yields a function that is continuous away from its cut and has a jump discontinuity of 2π across it.

The essential content is: no continuous single-valued argument function exists on all of $\mathbb{C} \setminus \{0\}$. This is a topological fact about the punctured plane, not an artifact of a particular convention.

The Argument and Winding Numbers

The winding number of a closed curve

Let $\gamma : [0, 1] \rightarrow \mathbb{C} \setminus \{0\}$ be a continuous closed curve (so $\gamma(0) = \gamma(1)$). As a point traverses γ , its argument changes continuously — but because γ is closed, the argument need not return to its starting value. It returns to a value differing from the initial value by an integer multiple of 2π . That integer is the **winding number** of γ about the origin:

$$\text{wind}(\gamma, 0) = \frac{1}{2\pi} [\theta(1) - \theta(0)], \quad (14)$$

where $\theta : [0, 1] \rightarrow \mathbb{R}$ is any continuous function satisfying $\gamma(t) = |\gamma(t)| e^{i\theta(t)}$. The existence of such a θ is guaranteed by the path-lifting property of covering spaces (see below). The integer $\text{wind}(\gamma, 0)$ counts the net number of counterclockwise loops γ makes around the origin.

The integral formula

In complex analysis, the winding number has the integral representation

$$\text{wind}(\gamma, 0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{1}{z} dz. \quad (15)$$

Writing $z = r e^{i\theta}$, one has $dz = (dr + ir d\theta) e^{i\theta}$, so $\frac{dz}{z} = \frac{dr}{r} + i d\theta$. Since γ is closed, the total change in $\ln r$ is zero, and the integral reduces to i times the total change in θ , which is $2\pi i$ times the winding number. This is why (15) is an integer.

More generally, the winding number about an arbitrary point $a \notin \gamma$ is

$$\text{wind}(\gamma, a) = \frac{1}{2\pi i} \oint_{\gamma} \frac{1}{z - a} dz. \quad (16)$$

Why the argument is a winding number

The multi-valued argument $\arg(z)$ can be understood as follows. Fix a base point $z_0 \in \mathbb{C} \setminus \{0\}$ and a value $\theta_0 \in \arg(z_0)$. For any other point z_1 , choose a path γ from z_0 to z_1 in $\mathbb{C} \setminus \{0\}$, and define the argument of z_1 along that path as θ_0 plus the total change in angle along γ . Different paths from z_0 to z_1 give different values, and two such values differ by 2π times the winding number of the closed loop formed by traversing one path forward and the other backward. The set of all attainable values is exactly the coset (3).

This is the fundamental reason for the multi-valuedness: the punctured plane $\mathbb{C} \setminus \{0\}$ has a nontrivial fundamental group,

$$\pi_1(\mathbb{C} \setminus \{0\}) \cong \mathbb{Z}, \quad (17)$$

generated by any simple loop around the origin. The integer labeling each homotopy class is the winding number. The argument function "detects" this topology.

The Riemann Surface of the Argument

The problem

The argument is multi-valued on $\mathbb{C} \setminus \{0\}$, and every branch cut produces a discontinuity. The Riemann surface resolves this by constructing a new domain on which the argument becomes a genuine (single-valued, continuous) function.

The universal cover of the punctured plane

The punctured plane $\mathbb{C} \setminus \{0\}$ is topologically an annulus, or equivalently the product $\mathbb{R}_{>0} \times S^1$ via polar coordinates $(r, e^{i\theta})$. Its universal covering space is

$$\widetilde{\mathbb{C} \setminus \{0\}} \cong \mathbb{R}_{>0} \times \mathbb{R}, \quad (18)$$

with covering map

$$p : \mathbb{R}_{>0} \times \mathbb{R} \rightarrow \mathbb{C} \setminus \{0\}, \quad p(r, \theta) = r e^{i\theta}. \quad (19)$$

The fiber over any point z is the set $\{(|z|, \theta) : \theta \in \arg(z)\}$, which is a discrete infinite set — one copy for each integer multiple of 2π that can be added. The group of deck transformations is

$$(r, \theta) \mapsto (r, \theta + 2\pi n), \quad n \in \mathbb{Z}, \quad (20)$$

which is isomorphic to $\mathbb{Z}$, matching the fundamental group (17).

The argument as a coordinate on the cover

On the covering space $\mathbb{R}_{>0} \times \mathbb{R}$, the argument is simply the projection onto the second factor:

$$\widetilde{\arg}(r, \theta) = \theta. \quad (21)$$

This is a perfectly well-defined, single-valued, continuous (indeed smooth) function. The multi-valuedness of $\arg$ on $\mathbb{C} \setminus \{0\}$ is an artifact of collapsing the covering space back to the base: each point of $\mathbb{C} \setminus \{0\}$ has infinitely many preimages under p , each labeled by a different value of θ .

Visualizing the Riemann surface

The covering space (18) can be visualized as a **helicoid** (infinite spiral staircase) sitting over the punctured plane. Each "sheet" corresponds to a particular interval of length 2π for θ . Moving continuously around the origin causes one to pass from one sheet to the next — the angle increases by 2π per revolution, and one never returns to the same sheet. This is the geometric content of the winding number: the number of sheets traversed equals the winding number of the path.

A branch cut in the base plane corresponds to slitting the helicoid along a vertical half-line and selecting a single sheet. The discontinuity of the branch at the cut corresponds to the jump from the top edge of one sheet to the bottom edge of the adjacent sheet.

The logarithm lives on the same surface

Since $\log z = \ln |z| + i \arg(z)$, the complex logarithm is also multi-valued for the same reason, and it becomes single-valued on the same covering space:

$$\widetilde{\log}(r, \theta) = \ln r + i\theta. \quad (22)$$

This is a biholomorphism from $\mathbb{R}_{>0} \times \mathbb{R}$ (viewed as a subset of $\mathbb{C}$ via $(r, \theta) \mapsto \ln r + i\theta$) onto all of $\mathbb{C}$. The covering map p is then $p = \exp \circ \widetilde{\log}^{-1}$, and the entire structure reduces to the statement that $\exp : \mathbb{C} \rightarrow \mathbb{C} \setminus \{0\}$ is the universal covering map of the punctured plane.

Summary of the Structural Content

The argument of a complex number encodes the angular position in the plane. Its multi-valuedness is not a defect but a reflection of the nontrivial topology of $\mathbb{C} \setminus \{0\}$: the fundamental group $\pi_1(\mathbb{C} \setminus \{0\}) \cong \mathbb{Z}$ measures the distinct homotopy classes of loops, each distinguished by a winding number, and the argument function is the map that reads off this winding number. The Riemann surface of the argument — the universal cover of the punctured plane — resolves the multi-valuedness into a single-valued function by "unwinding" the plane into an infinite helicoid, on which the argument is a global coordinate. Branch cuts are cross-sections of this covering, and every identity involving Arg modulo 2π is a shadow of an exact identity on the covering space.

